

Introduction and Python Basics

BMEG3105 Tutorial 1

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BMEG3105: Data analytics for personalized genomics and precision medicine

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What's the tutorial course for?

- Bridge the gap between Biology background and CS/math knowledge.
 - ▷ Most of you have not taken a CS course. We start from zero so later lectures are not blocked by notation or code.
- Specifically, introduce how to program and **how to learn programming**.
 - ▷ Remembering syntax is not the goal. You need a way to read an error, check a tiny example, and look up the rest.
- It helps you directly on the course project, because the project requires programming.
 - ▷ The programming assignments and the project all ask you to run and change a notebook, not only write a report.

What has changed compared to last year?

In the past, we went through the most useful programming language in both biology and data analytics, Python.

- Python basic grammar (fundamental tools)
 - ▷ Variables, lists, dictionaries, `if`, `for`, and functions: enough to read a short script.
- Useful Python packages such as NumPy, PyTorch, Biopython (advanced tools)
 - ▷ NumPy for matrices, PyTorch for tensors, Biopython for sequences. These show up again in lectures and programming assignments.
- Many cases and practices
 - ▷ Last year this took many weeks of drills. That block is now much shorter.

What has changed compared to last year?

Now, students can handle any course that needs programming by telling an AI agent:

```
Please help me finish {homework_pdf_path}.
```

- A full score on such homework proves whose ability?
 - ▷ The score may belong to the model. We still ask whether *you* can explain the result and catch a mistake.
- If AI helps you finish your bachelor's degree, what have you gotten from it?
 - ▷ If an agent writes every line, the degree no longer shows what you can do when the agent is wrong.

What has not changed?

- Knowing what exists in the world of computing and where to find answers
 - ▷ You do not memorize every function. You need the name of the idea (alignment, clustering) and where a tool lives.
- The readiness to bear the consequences of a decision
 - ▷ A leaked test set or a wrong metric is still your submission. The agent will not sit the exam for you.
- The ability to organize people, tools, and feedback to ultimately bring a project to fruition
 - ▷ A project is a plan, libraries, questions to TAs, and a revision after comments—not a single prompt.

How to bring a project to fruition in 2026?

1. Understand the fundamental logic of what you have done. **(Math)**
 - ▷ Alignment scores, distances, loss: if you cannot write the idea on paper, you cannot tell whether the code is right.
2. Use your hands to build tools for such logic. **(Programming)**
 - ▷ The formula has to become a notebook that runs. You may use an agent, but the pipeline is still yours.
3. Display the result and show your insight. **(Presentation)**
 - ▷ A number without a figure and a sentence is not enough. You have to say what it means.

Of course, you still need to understand every step; dirty work can be handed over to the agent, allowing human attention to focus more on the problem, design, and verification.

What do you need from us (our solution)

- We will give you very basic Python programming tutorials.
 - ▷ Three short notebooks: grammar, packages, and practice on biological data.
- We will show you how to learn code-related things by yourself.
 - ▷ Read the last line of an error, print a value, and test a case you can compute by hand.
- We will show you how to learn with the help of AI.
 - ▷ Later tutorials: how to write a prompt, and how to refuse an answer that does not check out.

Current plan

May adjust according to your feedback.

- **Python programming** (3–4 tuts): to make sure you have basic knowledge of Python
 - ▷ First weeks of term, so you can open Colab and follow a notebook before the programming assignments.
- **AI Agent basics** (2 tuts): how to correctly use an agent with the basic knowledge
 - ▷ After you can read Python, so you have something to check the agent against.
- **AI Agent for research projects** (3–4 tuts): how to build a research project using agents
 - ▷ Aimed at your project: data, experiment, and a report you can defend.

Tutorial 1: Python Basics

[Open the Colab notebook](#)

▷ File → Save a copy in Drive first if you want to edit. Leave the short problems as comments; we will add answers weekly.